

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)

Assessment Tool Weightage: 30%

Total Marks: 100

Time: 90 Mins

No. of Questions: 1

Annexure A Research Review & Analysis

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	
11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE QUESTION

S.No.	Research Review & Analysis Question	Primary Topic
1	Decision Tree Learning: Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.	Decision Trees
2	Neural Networks vs Statistical Learning: Review and compare research on neural-network and statistical-learning approaches for prediction and classification. Analyze their performance, data requirements, generalization, interpretability, computational cost, and suitability for different applications.	Statistical Learning & Neural Networks
3	Kernel Methods: Analyze research on SVM and kernel methods for non-linear classification. Compare linear, polynomial, and RBF kernels and evaluate their advantages and limitations compared with neural-network-based approaches.	Kernel Methods
4	Reinforcement Learning: Review research on Q-Learning, SARSA, and Deep Q-Networks. Compare their learning mechanisms, exploration strategies, convergence, computational requirements, and real-world applications. Identify challenges and research gaps.	Reinforcement Learning
5	Explainable and Generalizable AI: Review research on Inductive Learning and Explanation-Based Learning and analyze how these approaches contribute to generalization and explainability in AI. Identify current limitations and propose future research directions.	Inductive & Explanation-Based Learning

Structure of the Report:

S.No.	Paper Section	Expected Content
1	Title	Specific and concise title related to the selected research question/topic.
2	Abstract	Brief summary of the research problem, review methodology, major findings, research gap, and conclusion.
3	Keywords	4–6 important keywords related to the selected topic.
4	1. Introduction	Introduce the AI topic, background, importance, motivation, and scope of the review.
5	2. Research Problem and Objectives	Clearly state the research problem and specific objectives of the review.
6	3. Literature Review	Review relevant research papers and summarize major approaches, methodologies, findings, and contributions.
7	4. Comparative Analysis	Compare the reviewed approaches based on suitable parameters such as accuracy, complexity, interpretability, computational cost, scalability, and generalization.
8	5. Critical Analysis	Discuss strengths, weaknesses, limitations, assumptions, and practical challenges identified across the reviewed studies.
9	6. Research Gap	Identify unresolved problems, limitations in existing research, and areas requiring further investigation.
10	7. Proposed Research Direction	Suggest possible solutions, improvements, or future research directions based on the identified gap.
11	8. Discussion	Discuss the overall insights obtained from the review and their implications for AI research and applications.
12	9. Conclusion	Summarize the major findings, key insights, research gap, and future direction.
13	References	List all research papers, books, datasets, websites, and other sources using a consistent citation style.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.
3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.

4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.
6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.

7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.
10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic-integrity concerns.